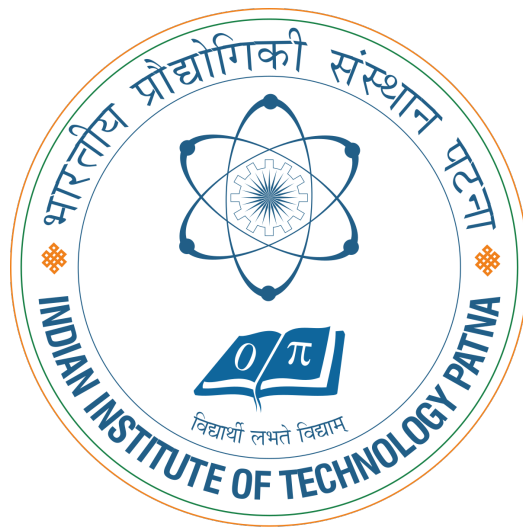


Multi-class classification using Fully Connected Neural Network



Ritik Yadav (2621MA03)

Sahnil Siddique (2621MA02)

Suman Acharya (2621MA11)

Department of Mathematics

Indian Institute of Technology Patna

1. Purpose

This report documents the implementation and experimental results contained in the submitted Jupyter notebook for Task 2. The experiment compares three fully connected neural-network architectures and six optimization methods for five-class Fashion-MNIST classification. The reported numerical results are taken directly from the notebook outputs.

2. Selected classes

Encoded	Fashion-MNIST class
0	T-shirt/top
1	Trouser
2	Pullover
3	Dress
4	Coat

3. Dataset and split used in the notebook

The notebook loaded the official Fashion-MNIST training set (60,000 images) and test set (10,000 images), then retained classes 0–4. This produced 30,000 selected training samples and 5,000 selected test samples. Images were normalized to $[0, 1]$ and flattened from 28×28 pixels to 784 input features. The selected training portion was stratified into 24,000 training samples and 6,000 validation samples; the 5,000 selected official test samples were kept as the final test set.

4. Experimental configuration

Item	Notebook setting
Input	784 features (28×28 flattened grayscale image)
Output	5 classes
Hidden-layer architectures	3, 4 and 5 hidden layers
Activation	ReLU in hidden layers
Loss	Cross-entropy loss
Learning rate	0.001
Momentum	0.9 for Momentum and NAG
RMSProp α	0.99
Adam β_1 , β_2	0.9, 0.999
ϵ	10nn
Random seed	42
Maximum epochs	200
Convergence tolerance	10nn change in training error
Hardware	CPU; PyTorch 2.11.0+cpu

5. Model Architectures and Optimizers

All models use the same 784-dimensional input and five-unit output layer. Hidden layers use fully connected Linear layers followed by ReLU activation. The three architectures used in the notebook were:

Architecture	Hidden-layer node counts
Architecture_3HL	$256 \rightarrow 128 \rightarrow 64$
Architecture_4HL	$256 \rightarrow 128 \rightarrow 64 \rightarrow 32$
Architecture_5HL	$256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$

The six optimizers compared were SGD, Vanilla GD, Momentum, NAG, RMSProp and Adam. For every architecture, the notebook created one initial model state and copied the same initial weights into each optimizer run, supporting a controlled optimizer comparison.

6. Convergence Epochs and Training Time

Architecture	Optimizer	Epochs	Time (s)
3HL	SGD	149	628.52
3HL	Vanilla GD	171	738.21
3HL	Momentum	76	339.87
3HL	NAG	112	502.11
3HL	RMSPProp	38	189.99
3HL	Adam	52	278.55
4HL	SGD	200	848.68
4HL	Vanilla GD	200	841.83
4HL	Momentum	77	346.20
4HL	NAG	116	555.22
4HL	RMSPProp	59	307.99
4HL	Adam	190	1051.42
5HL	SGD	200	860.25
5HL	Vanilla GD	200	855.05
5HL	Momentum	66	288.62
5HL	NAG	113	513.74
5HL	RMSPProp	109	573.76
5HL	Adam	35	188.45

A run is reported as converged when the absolute difference between consecutive average training errors falls below 10^{-6} , after the minimum epoch condition. Runs reaching 200 epochs are reported at the experiment's maximum-epoch cap rather than as converged earlier.

7. Final Training and Validation Accuracy

Architecture	Optimizer	Train Acc. %	Validation Acc. %	Epochs
3HL	SGD	88.28	88.77	149
3HL	Vanilla GD	88.57	88.90	171
3HL	Momentum	91.83	90.33	76
3HL	NAG	95.87	91.45	112
3HL	RMSProp	95.80	91.35	38
3HL	Adam	97.98	91.80	52
4HL	SGD	88.09	87.90	200
4HL	Vanilla GD	88.00	87.97	200
4HL	Momentum	94.24	91.55	77
4HL	NAG	95.14	89.58	116
4HL	RMSProp	97.60	91.62	59
4HL	Adam	99.59	91.88	190
5HL	SGD	86.97	87.42	200
5HL	Vanilla GD	86.74	87.17	200
5HL	Momentum	94.06	91.33	66
5HL	NAG	97.65	91.02	113
5HL	RMSProp	98.75	92.13	109
5HL	Adam	96.28	92.05	35

Architecture-level validation summary

Using the notebook's stated selection rule (highest final validation accuracy), Architecture_5HL with RMSProp was selected, with validation accuracy of 92.13%.

Architecture	Optimizer associated with highest final validation accuracy	Validation accuracy
3HL	Adam	91.80%
4HL	Adam	91.88%
5HL	RMSProp	92.13%

8. Training-Error Curves

The following figures reproduce the training-error plots stored in the submitted notebook. Each plot overlays the six optimizers for one architecture.

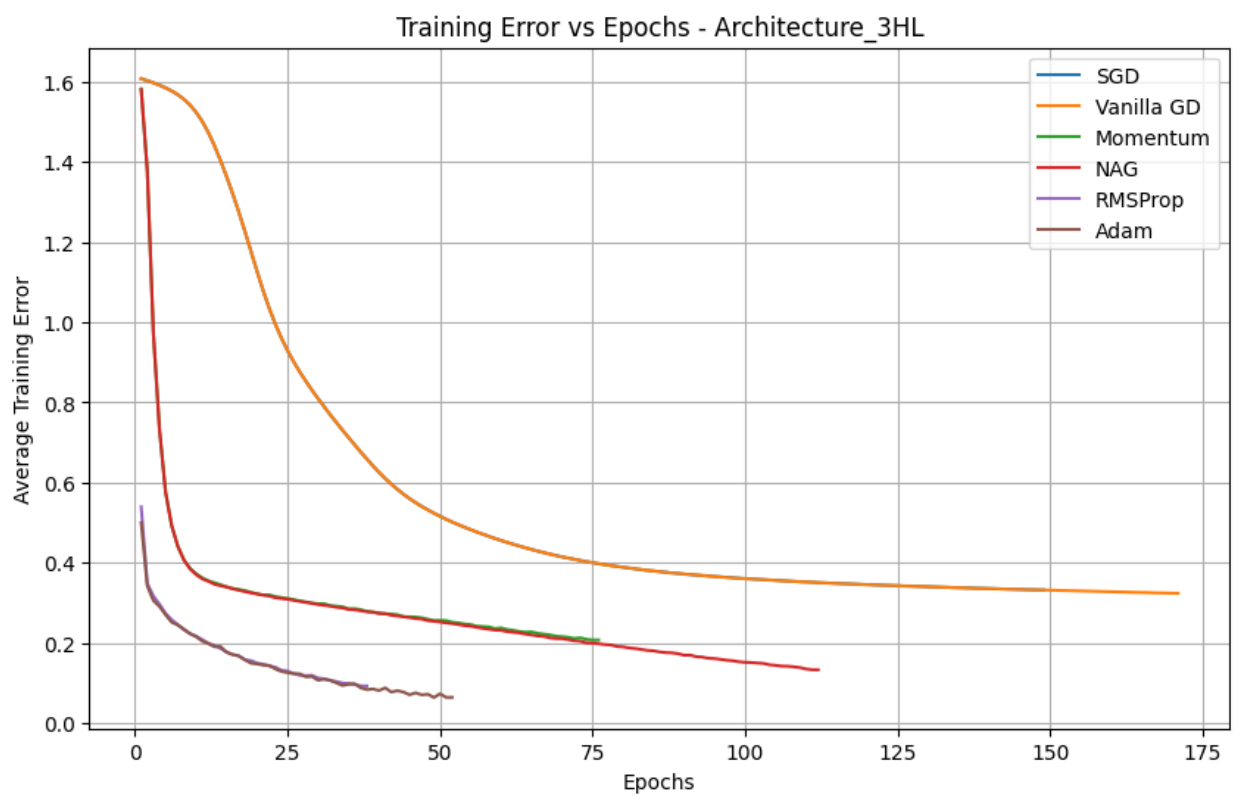


Figure 1: Average training error vs. epochs — Architecture_3HL

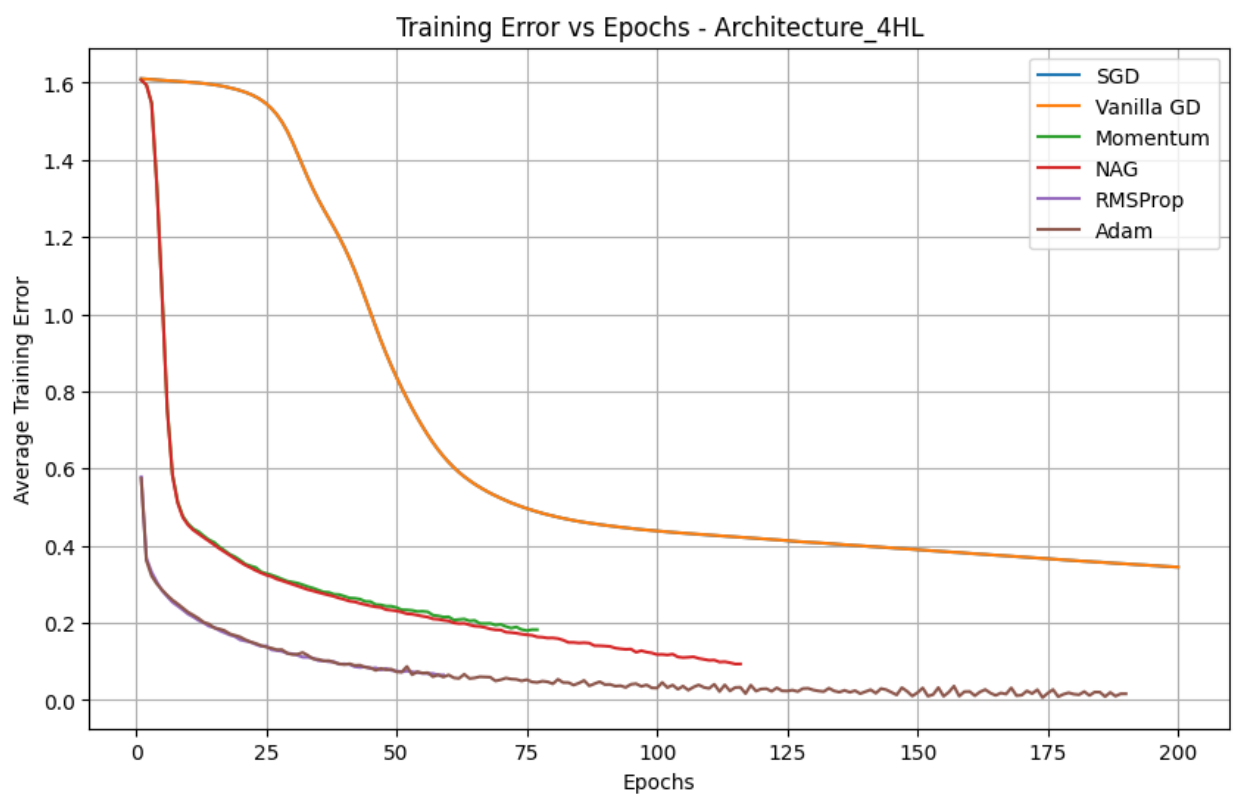


Figure 2: Average training error vs. epochs — Architecture_4HL

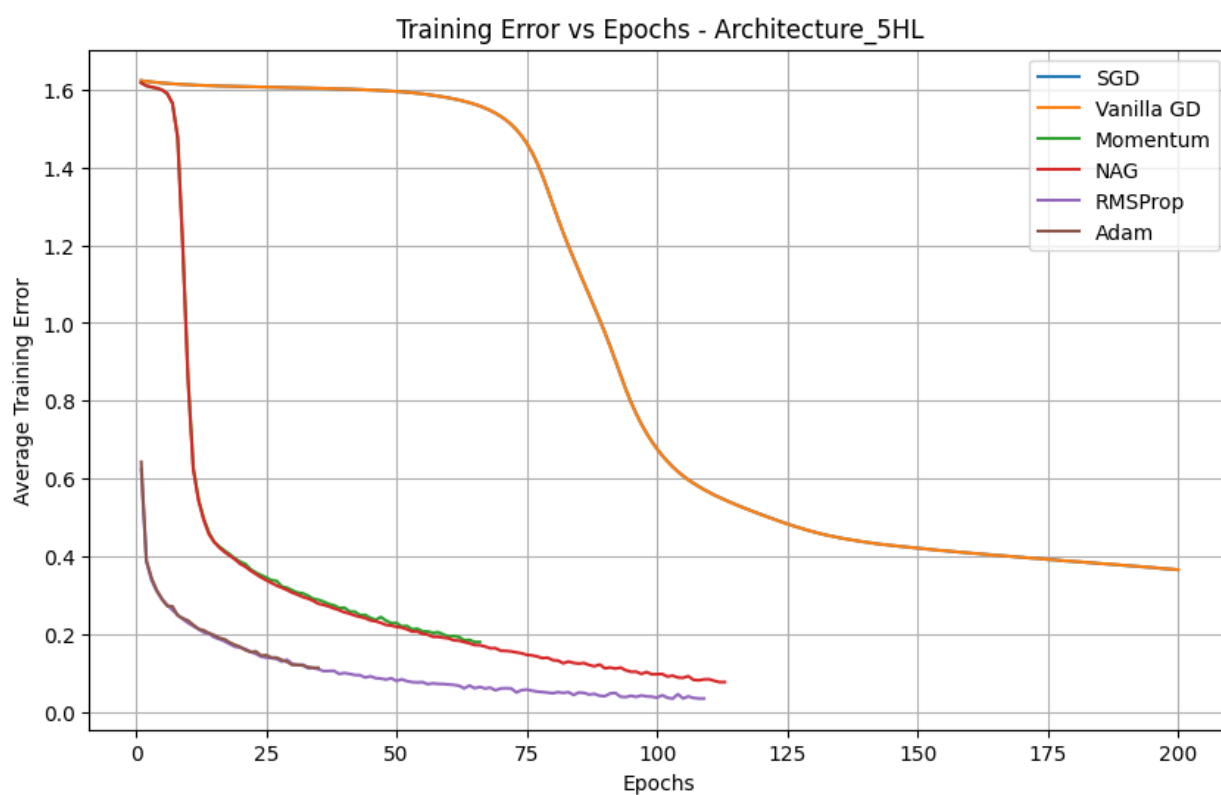


Figure 3: Average training error vs. epochs — Architecture_5HL

9. Validation Curves

Validation-error and validation-accuracy curves from the notebook are shown below.

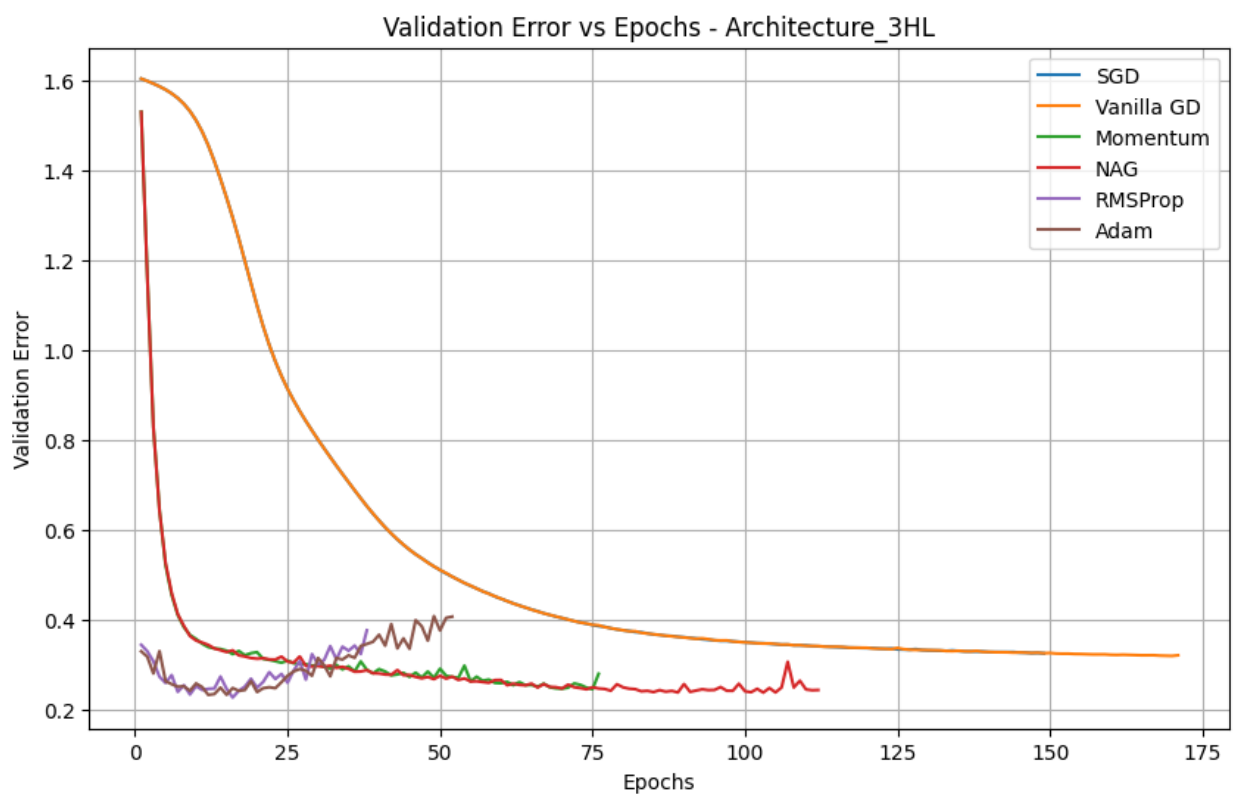


Figure 4: Validation error vs. epochs — Architecture_3HL

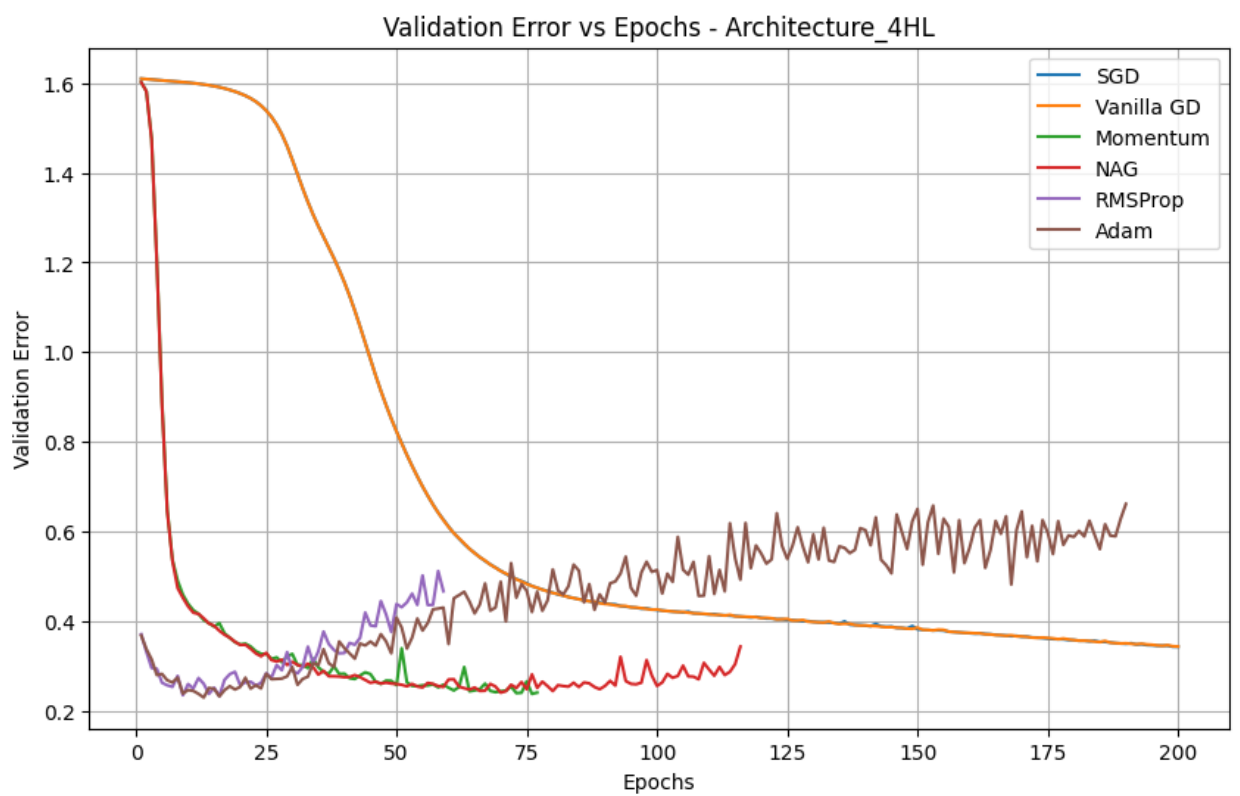


Figure 5: Validation error vs. epochs — Architecture.4HL

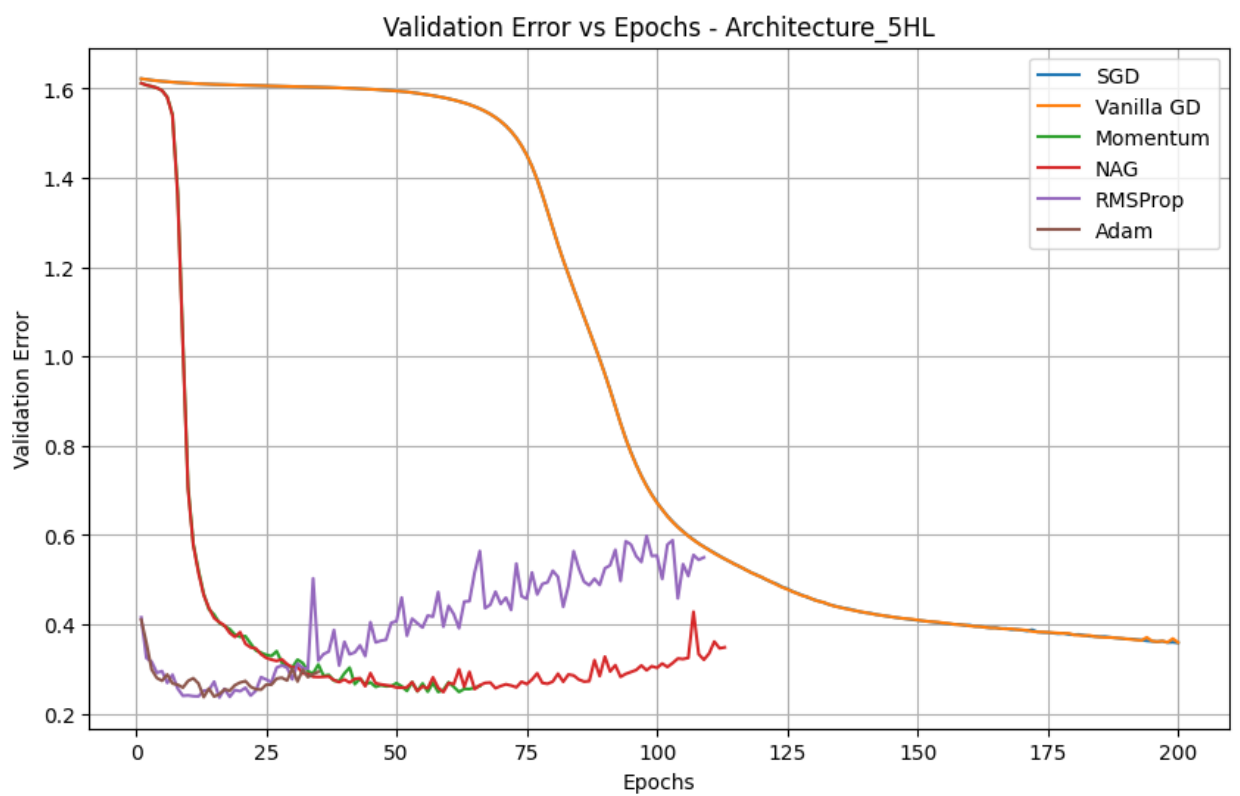


Figure 6: Validation error vs. epochs — Architecture_5HL

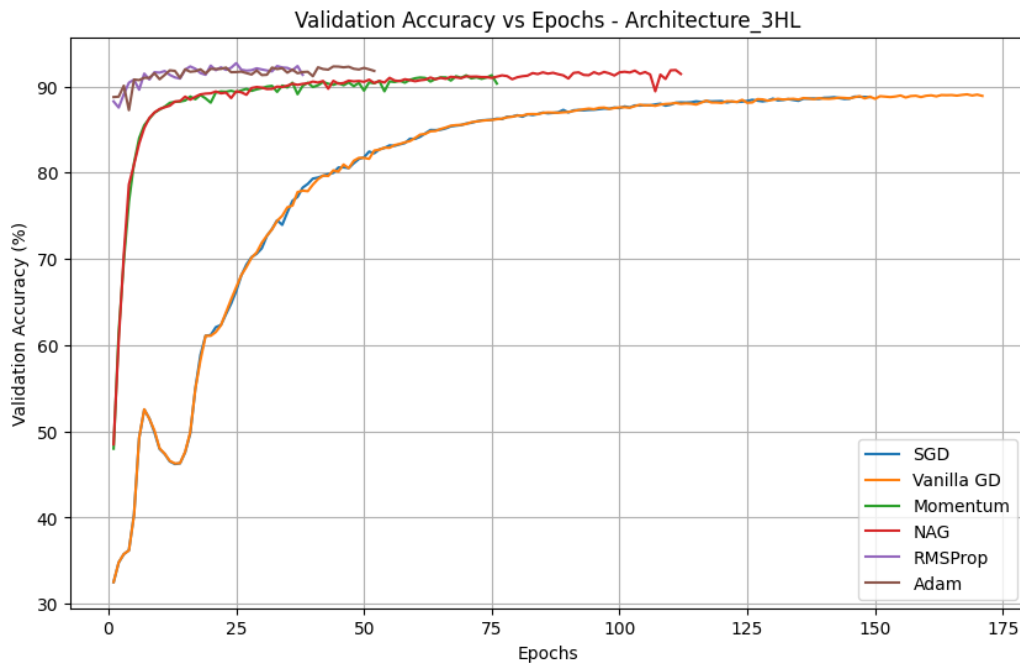


Figure 7: Validation accuracy vs. epochs — Architecture_3HL

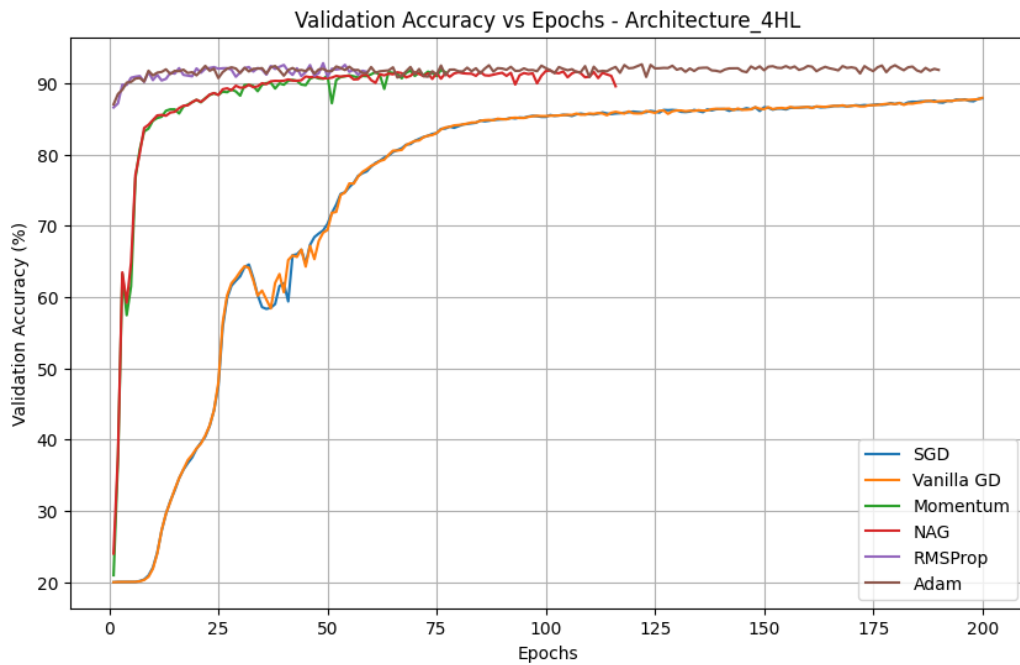


Figure 8: Validation accuracy vs. epochs — Architecture_4HL

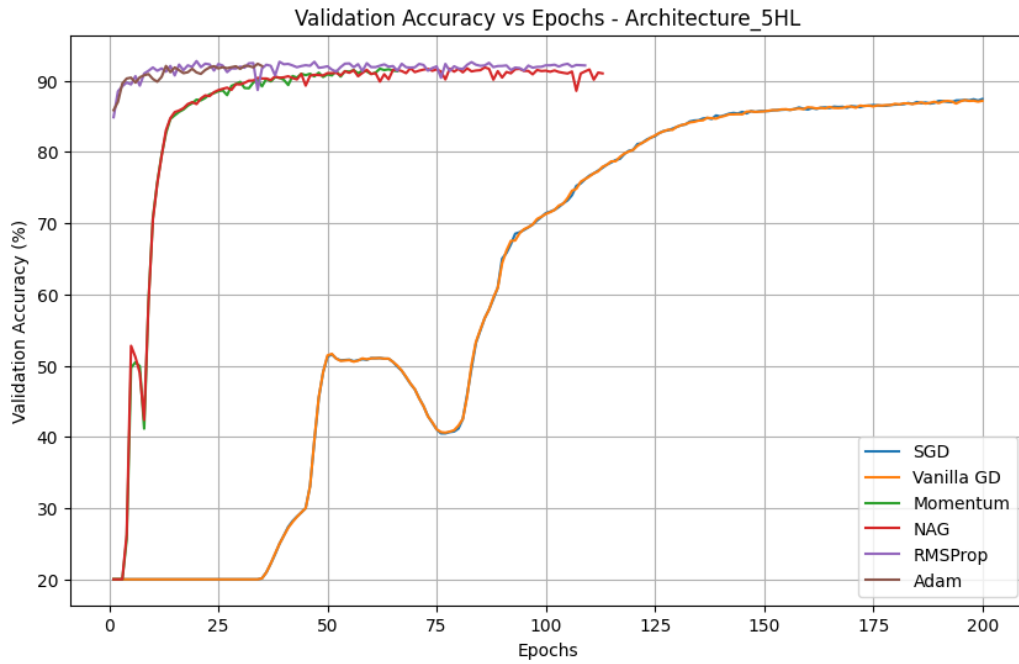


Figure 9: Validation accuracy vs. epochs — Architecture_5HL

10. Selected Model: Test Performance and Confusion Matrix

The selected model in the notebook is Architecture_5HL with RMSProp. Its final evaluation on the held-out selected test set produced a test accuracy of 90.80%.

Actual \ Predicted	T-shirt/top	Trouser	Pullover	Dress	Coat
T-shirt/top	937	1	20	35	7
Trouser	2	976	3	14	5
Pullover	26	0	842	10	122
Dress	25	6	13	900	56
Coat	3	0	89	23	885

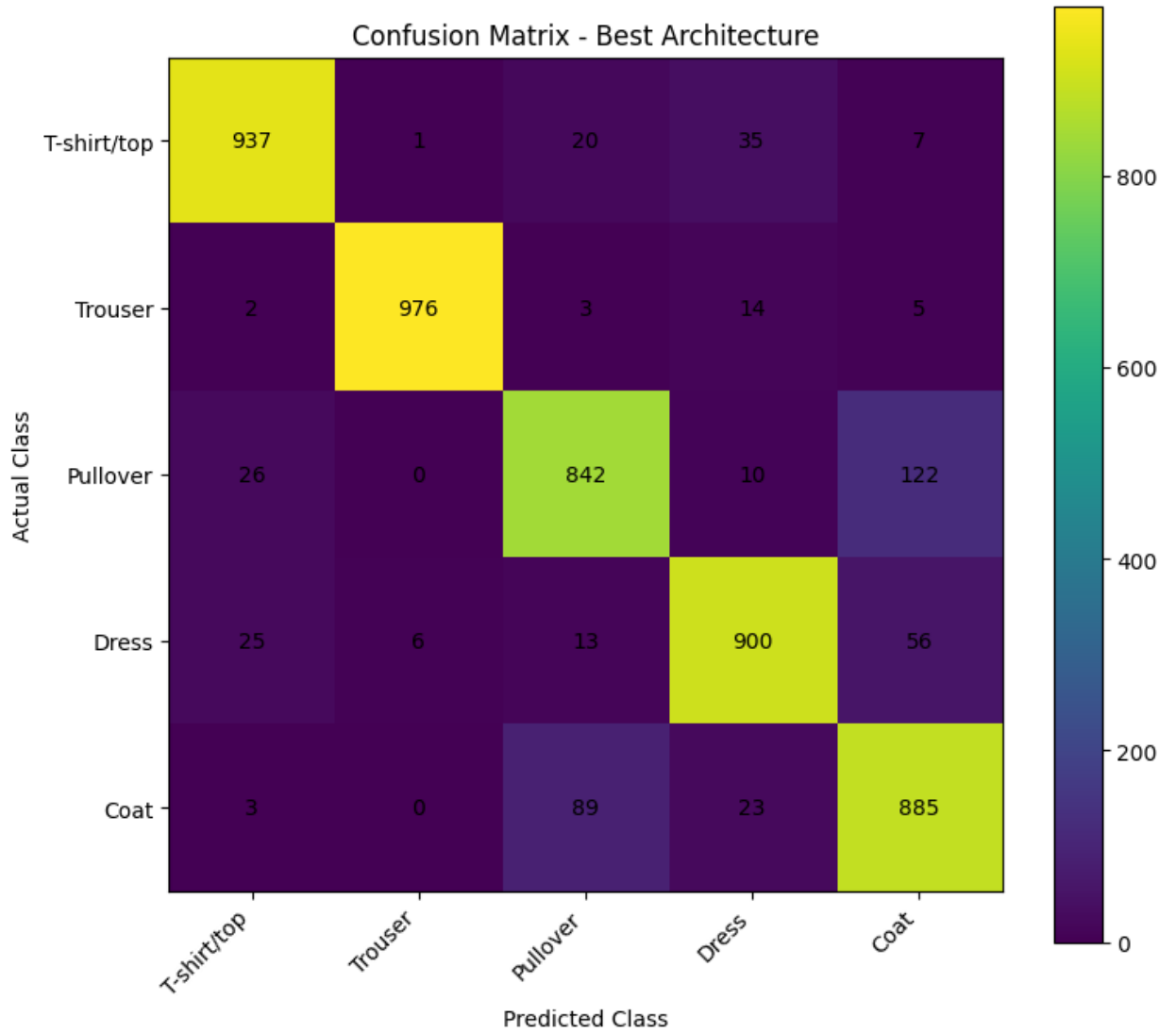


Figure 10: Confusion matrix for the selected Architecture_5HL + RMSProp model.

Per-class accuracies reported by the notebook were: T-shirt/top 93.70%, Trouser 97.60%, Pullover 84.20%, Dress 90.00%, and Coat 88.50%. The largest visible confusion in the matrix is between Pullover and Coat: 122 Pullover examples were predicted as Coat, while 89 Coat examples were predicted as Pullover.

11. Discussion of Results

- **Effect of optimization:** The convergence behavior differs substantially across optimizers. In the 3-hidden-layer architecture, RMSProp reached the stopping criterion in 38 epochs, while Vanilla GD required 171 epochs. In the 5-hidden-layer architecture, Adam reached the stopping criterion in 35 epochs, while SGD and Vanilla GD reached the 200-epoch cap.
- **Validation performance:** The best final validation accuracy recorded within each architecture was 91.8091.88% for 4HL, and 92.13% for 5HL.
- **Training versus validation:** Several runs have training accuracy substantially above validation accuracy, especially deeper networks with Adam or RMSProp. This indicates a gap between fitting the training data and generalization to the validation set.
- **Class-level behavior:** Trouser has the highest reported per-class accuracy (97.60%), while Pullover has the lowest (84.20%). The confusion matrix shows that Pullover and Coat are comparatively difficult to distinguish for this fully connected model.
- **Reproducibility:** The notebook uses random seed 42 and copies the same initial state to optimizer runs within each architecture, making the optimizer comparison more controlled.

12. Implementation Note

There are two important differences between the assignment specification shown in the supplied task sheet and the submitted notebook implementation. First, the notebook uses a batch size of 100 for the training DataLoader for all optimizer runs; the task sheet specifies `batch_size=1` for SGD, Momentum, NAG, RMSProp and Adam, and a full-batch setting for vanilla gradient descent. Second, the notebook filters the official Fashion-MNIST train and test sets separately and then performs an 80%/20% stratified training/validation split on the selected training set, resulting in 24,000 training, 6,000 validation and 5,000 test samples. The task sheet describes a 64%/16%/20% split with respect to the complete selected dataset. Therefore, the numerical results in this report describe the notebook as executed and should not be presented as a strict reproduction of the task-sheet split/batch-size requirements.

13. Conclusion

The notebook successfully implements a five-class Fashion-MNIST classification experiment using fully connected neural networks with 3, 4 and 5 hidden layers and six optimizers. Across the recorded experiments, the 5-hidden-layer architecture paired with RMSProp produced the highest final validation accuracy recorded in the notebook (92.13%) and achieved 90.80% test accuracy. The confusion matrix indicates strong recognition of Trouser and T-shirt/top, with greater confusion between Pullover and Coat. The results also show that optimizer choice has a substantial effect on convergence speed and final validation performance.